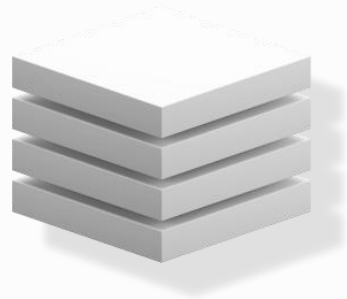




Stock Price Prediction with **Cross-Firm Integration**

Minseo Kim, Jihong Min, Dayoon Lee



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Baseline

GOAL: CAPTURING "EXTERNAL MARKET CONDITIONS"

Traditional price-only models fail to reflect external factors (News, Macroeconomics).

The key challenge is: How do we capture these efficiently?



Recent Trend: NLP-Based

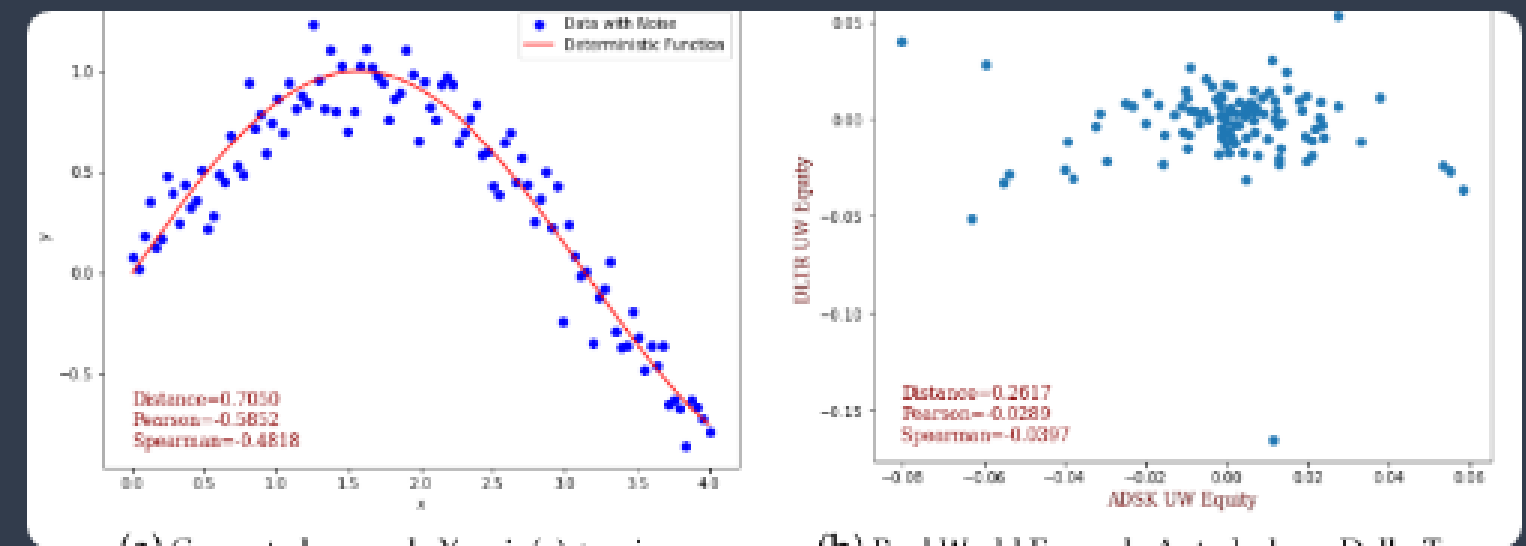


Attempts to capture market conditions directly by processing massive amounts of news text.

Limitation: High Cost & Resource Intensive



Our Approach: Interconnectedness



Captures market conditions indirectly by analyzing how external shocks ripple through the network.

Solution: Efficient Proxy for External Factors

Related Works

NLP, multimodal based

<LLM-Augmented Linear Transformer–CNN for Enhanced Stock Price Prediction> (Zhou et al., 2025)

→ (-): **costly, requires extra resources like financial text.**

Interconnectedness, Spillover

<Stock Returns' Co-Movement: A Spatial Model with Convex Combination of Connectivity Matrices> (Abdallah et al., 2025)

- cross- and within-country linkages between different stocks

<Comparative Analysis of Spillover Effects in the Global Stock Market under Normal and Extreme Market Conditions> (Liu et al., 2024)

- volatility spillover index to model interconnectedness in 20 major stock markets.

→ (X) **use identified comovement patterns, spillover effects in prediction.**

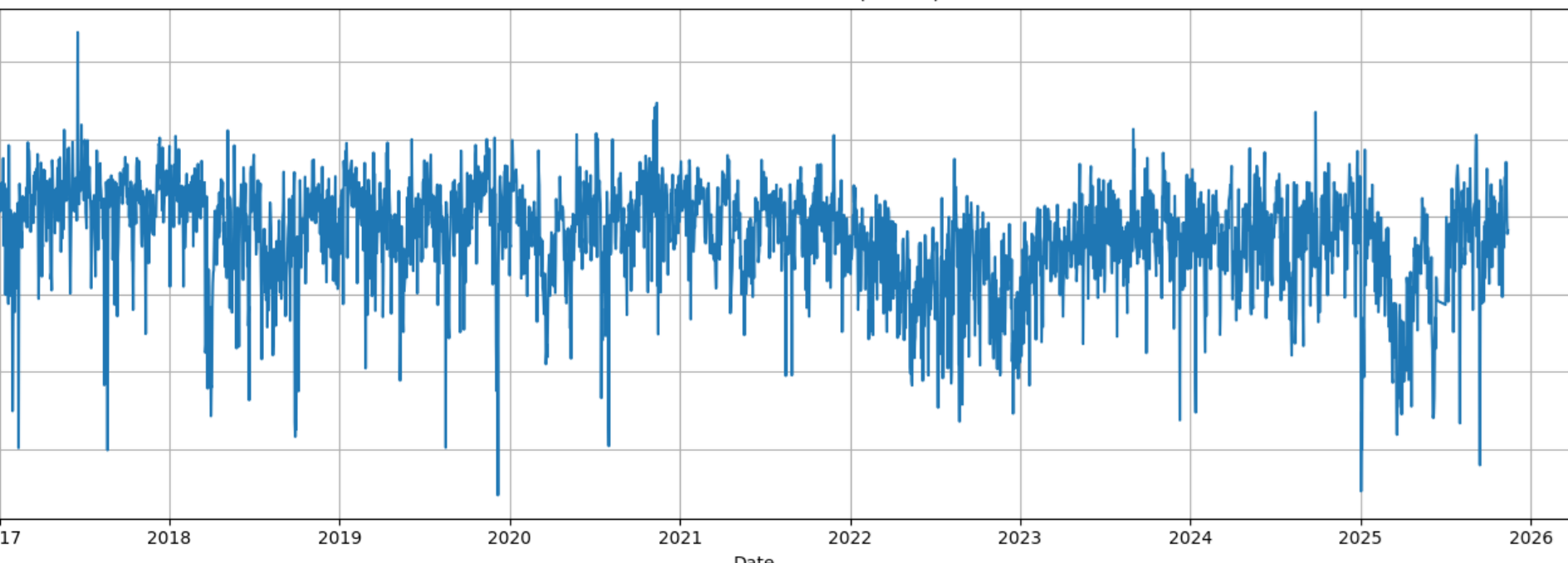
→ (X) **comparison between nlp-based vs interconnectedness-based**

Approach

Sentiment-based

- GDELT Tone score based on global news articles

Tesla News Tone Over Time (GDELT)



Correlation, interconnectedness-based

- Pearson correlation, CCF(Cross-Correlation Function), DTW similarity score, Granger causality test
- Find correlated features (c.f: log-return, volatility)
→ VARMA/LSTM

EDA



01

Rapid increase, Rapid drop

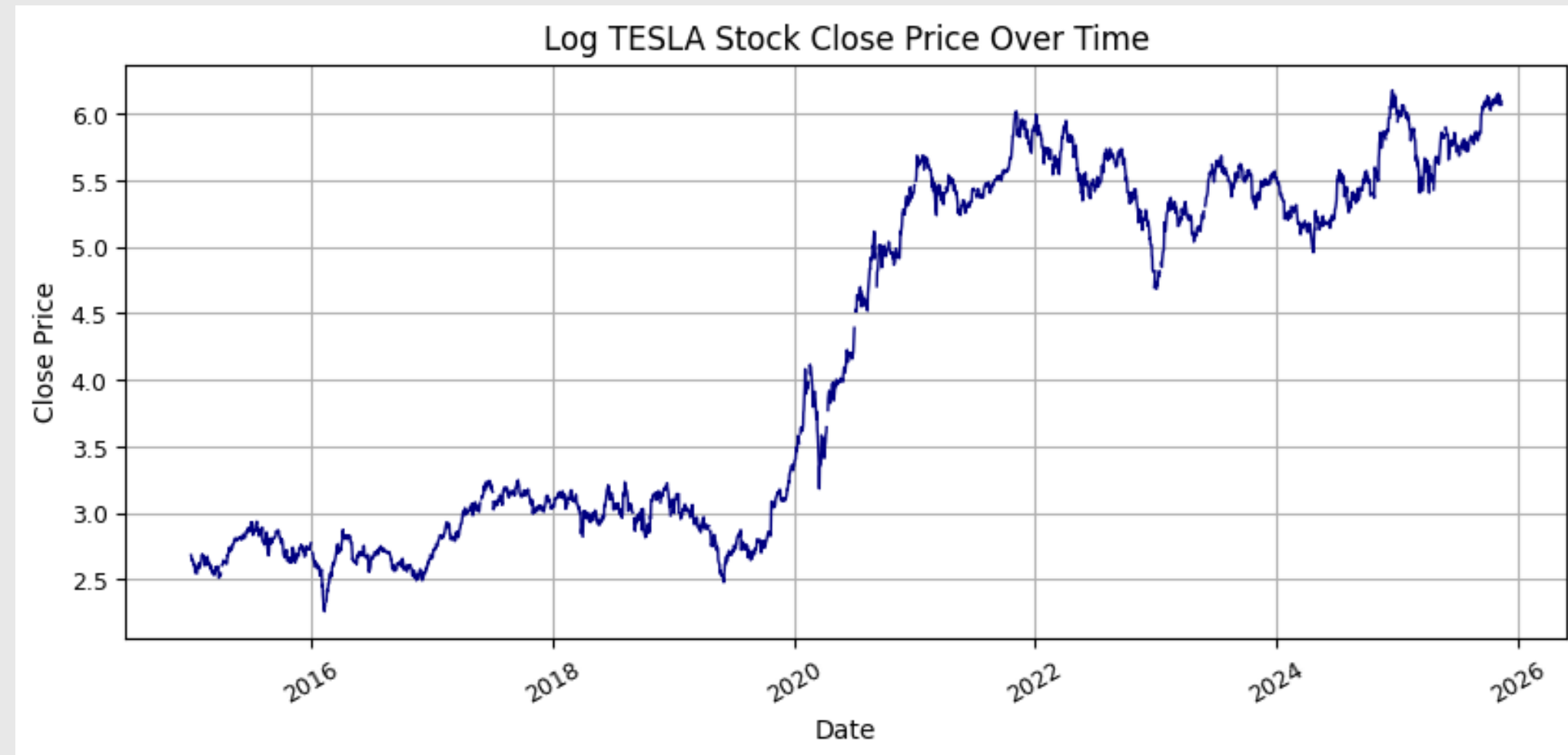
02

Missing value check

03

Outlier Check

Log transformation



1

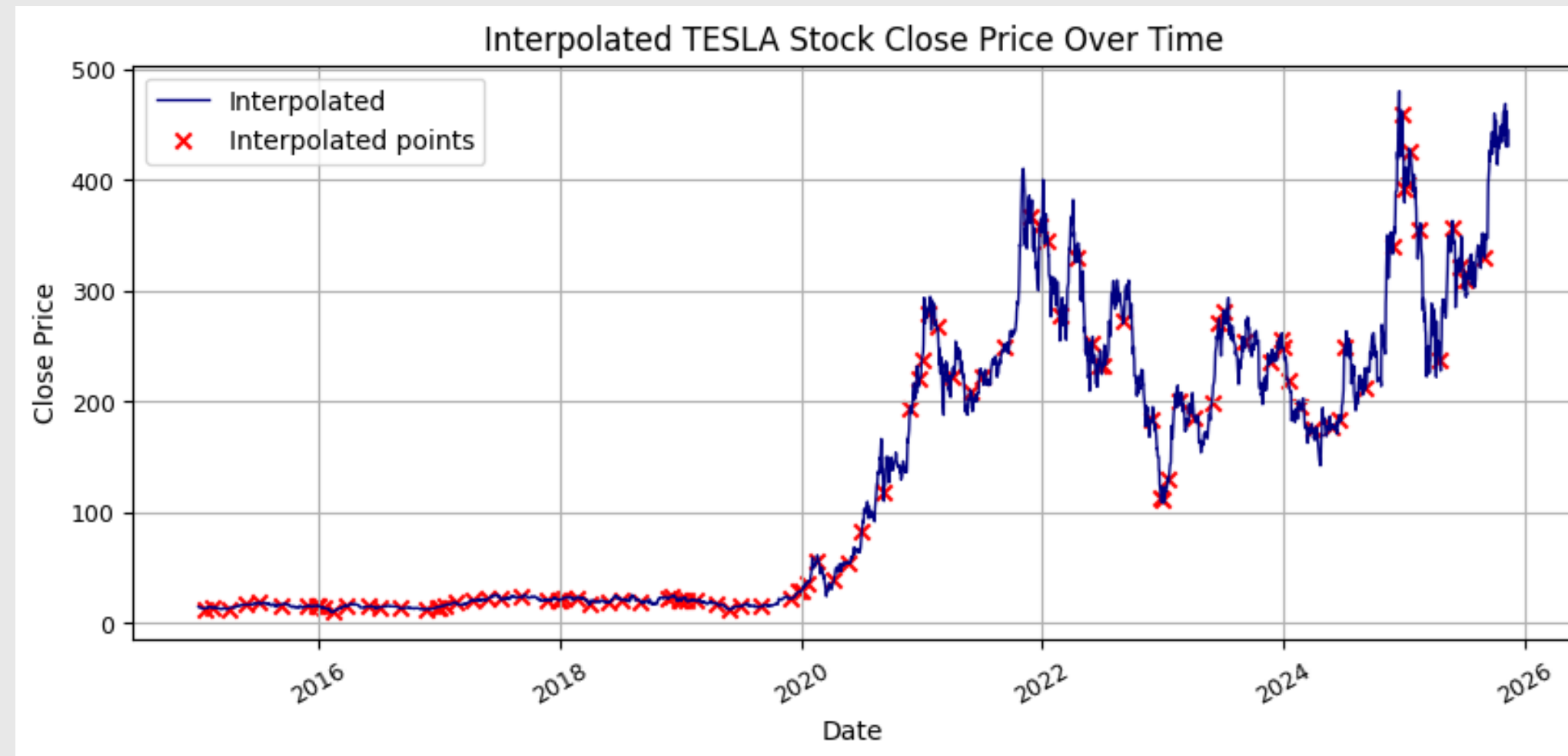
Apply log transformation to adjust scale

Missing Value



Missing Value occurs during the process of aligning domestic and international companies

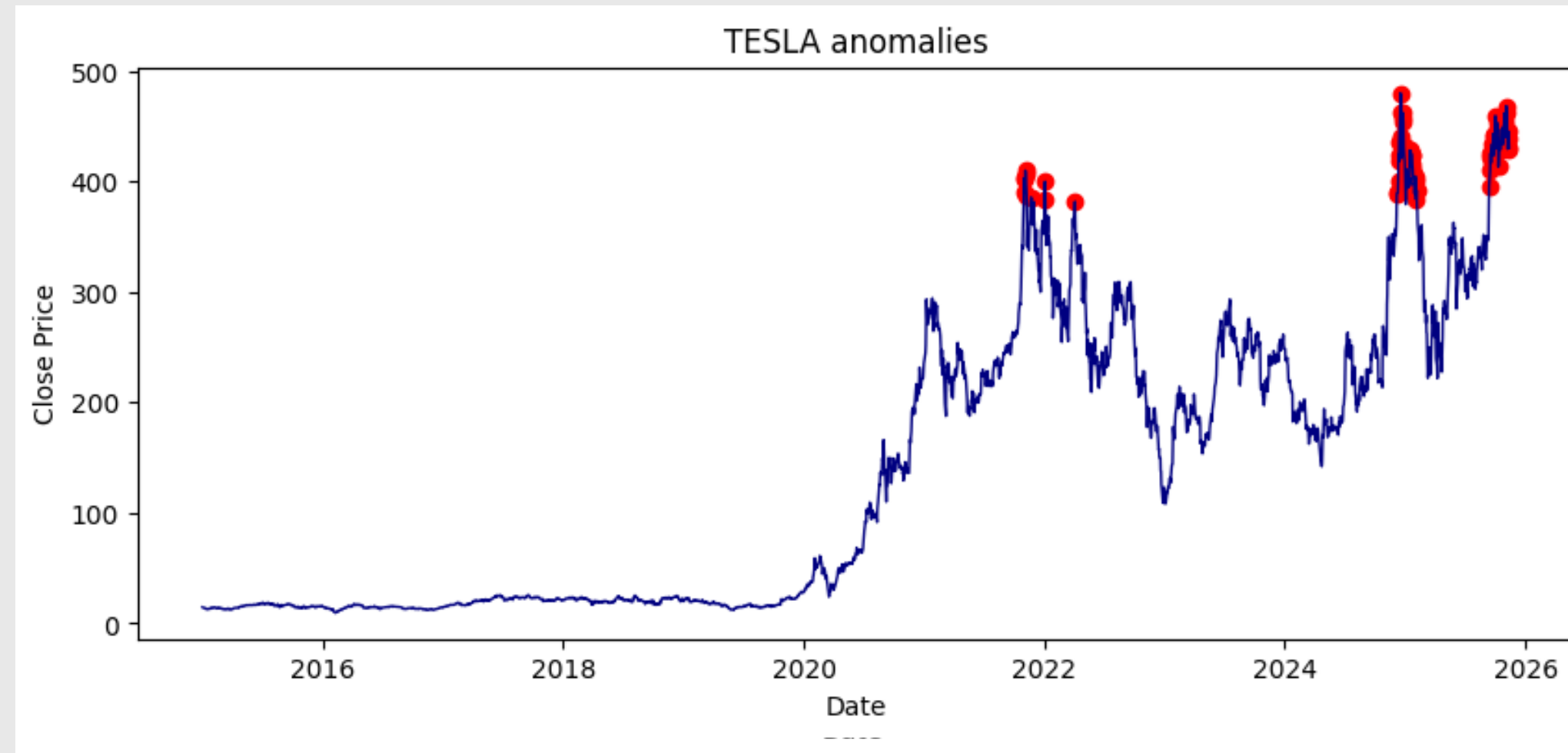
Missing Value



1

101 interpolated points follows original trends with interpolation

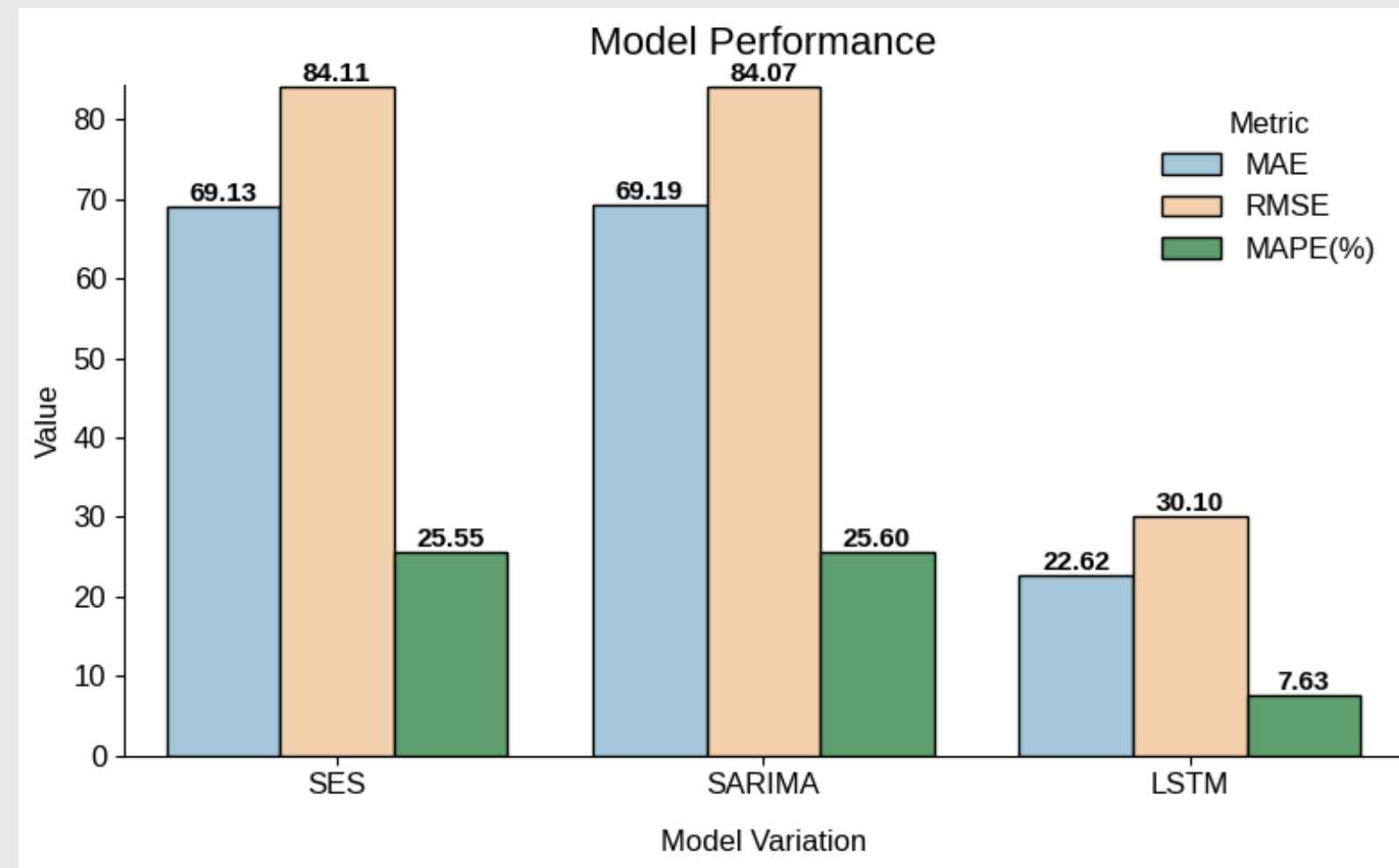
Outlier check



1 Prices after the sharp increase are considered as outliers when detecting with 3σ

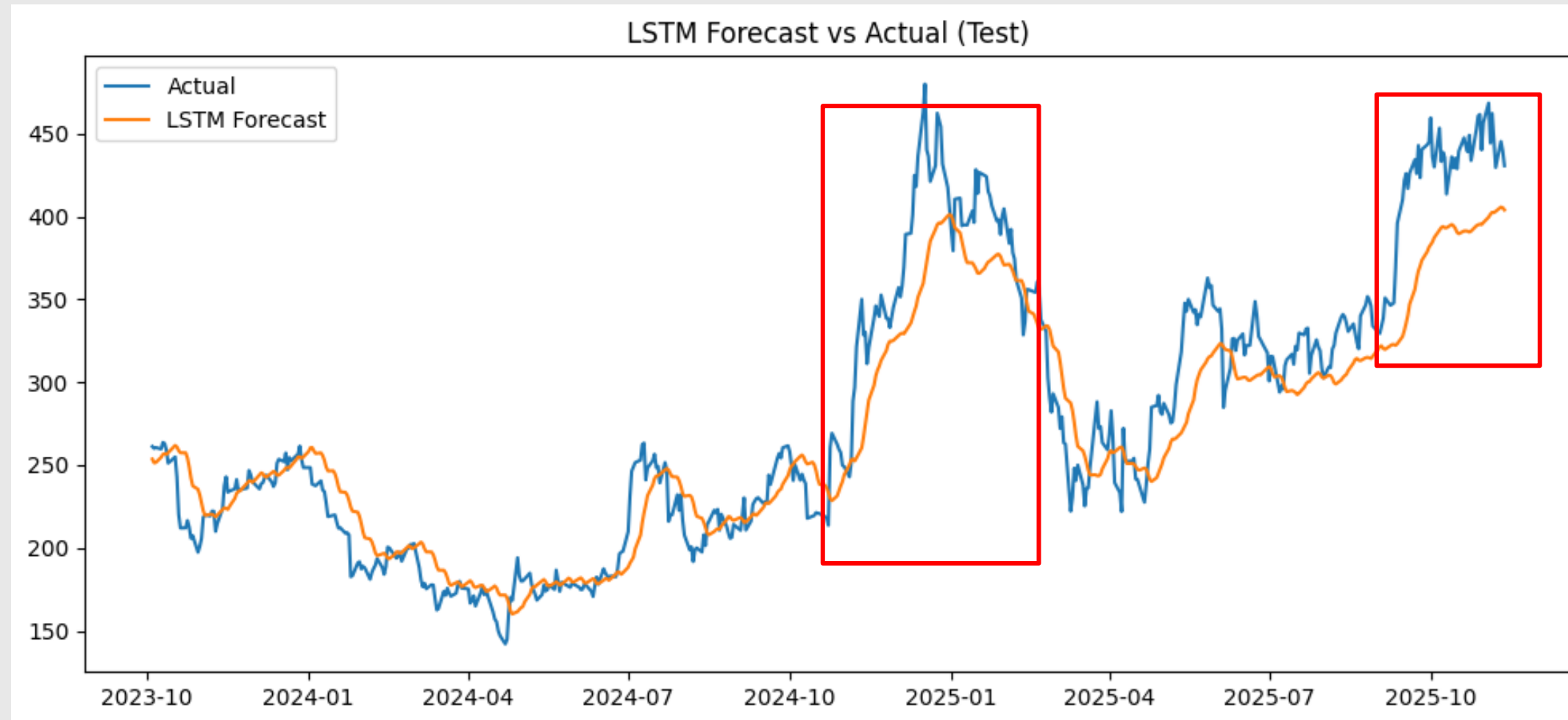
However, these values reflect characteristics of TESLA data → preserve them

Baseline Performance



1 LSTM capture variations relatively well and achieves moderate performance

Baseline Performance



However, there's still gap between prediction and real prices

Baseline Performance

Tesla CEO Elon Musk unveils 'Cybercab' robotaxi

By Abhirup Roy and Akash Sriram

October 11, 2024 5:18 PM GMT+9 · Updated October 11, 2024



2024 October

Global EV sales growth hits 23% in October, research firm says

By Amir Orusov and Jesus Calero

November 12, 2025 9:17 AM GMT+9 · Updated November 12, 2025



2025 November

1

Fails to handle sharp increase caused by [external](#) events.

Cybercab release: Cannot reflect with hyundai data

EV market expansion: Can reflect with hyundai data

Summary

01

Naive approach fail to capture variation caused by external events

02

NLP-based approach suggested by previous approach are too costly

03

Check efficiency of interconnectedness-based approach
by comparing it to sentiment-based approach
